

Jerry Barillas

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Education

University of California, San Diego

B.S. in Cognitive Science (Design & Interaction), Minor in Computer Science

September 2023 - Expected June 2027

Relevant Coursework: Object-Oriented Programming, Scientific Computing, Systems Programming and Software Tools, Advanced Data Structures

Experience

UCSDelivery | ACM Hack Projects

December 2025 - January 2026

TypeScript, MariaDB, Docker, Express.js, Node.js, Railway

- Collaborated in a team to build a market delivery web app with React frontend and Typescript Node.js/Express.js backend, building HTTP-based REST APIs
- Implemented session-based authentication and role-based authorization for cart/order endpoints
- Integrated MariaDB with parameterized SQL queries and developed in a Docker containerized environment; supported deployment on Railway

Projects

TritonTrade

February 2026 - Present

Java, Spring Boot, Spring Data JPA, Flyway, PostgreSQL, Docker

- Scaffolded a Spring Boot REST API with clean package structure (controllers, entities, repositories)
- Provisioned PostgreSQL using Docker and configured Spring datasource settings for reliable local development
- Designed a listings schema with Flyway migrations and implemented a JPA listing entity plus JPA repository

Image Processing Toolkit

January 2024

Python, Pillow (PIL)

- Built a Python image processing toolkit using Pillow for reusable pixel-level transformations
- Implemented edge detection by computing grayscale intensity differences between adjacent pixels with threshold-based classification
- Packaged image functions for repeatable use across arbitrary input photos

Technical Skills

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL

Libraries/Frameworks: NumPy, Pandas, Matplotlib, Pillow

Developer Tools: Git/GitHub, VS Code, Unix, Docker, Railway, Flyway